

Phase3-Feedback: G1-s3

- [3.1] Actor Description: A- , reasonably written - check as a group
- [3.2] Overall Use-Case Diagram: B- , has several issues, check as a team and match with scenarios and described use cases
- [3.3] Described use-cases: C/B, have issues, check individually
- [3.4] Overall Activity Diagram: B-, but has some issues, check as a team.

=> Noted Names of the members of the team who have contributed to the team-tasks, on each of the team-tasks: Actors Descriptions, Overall Use Case diagram, Activity Diagram

Overall Mark/Evaluation:(B): needs Major improvements

Individual Mark: individual mark will be combined together and provided on all individual tasks in the final report [check feedback on your individual tasks]

If issues/comments, for team-tasks and individual-tasks, are addressed for the final report, tasks may get reconsidered.

Department
Software Engineering (COMP 433)
Instructor: Prof. Adel Taweel

Phase 3
Requirements Analysis and Modelling
Printing Shop
Prepared by Group No. 1 :

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A-

Task 3.1 — Actor Analysis

Lead: Razan Froukh

Contributors: Tala Saabneh (extraction), Dareen Abualhaj (reviewing), Taima Nazal (verification), Shahd Shwekyeyeh (formatting)

Actor	Semantics/Description
Customer	This is a primary human actor that represents a registered user who registers, selects printing options, places and pays for print orders, manages pending orders, submits inquiries, and rates completed orders. ambiguous Verb/Use cases: describe the actor from a business/system usage/use cases
Admin	This is a primary human actor who oversees the entire printing shop system. Their responsibilities include managing user accounts, monitoring system operations, reviewing business performance, and ensuring platform compliance with shop policies.
PrintOperator	This is a primary human actor responsible for processing print jobs after order confirmation. They handle printing, conduct quality checks, and package completed orders for delivery.
SupportStaff	This is a primary human actor responsible for handling customer inquiries and issue reports submitted through the system. They review incoming messages, respond to customers, and escalate unresolved issues to the admin.
FinanceStaff	This is a primary human actor who manages financial operations within the system. They handle payment records, process refunds, prepare invoices, and generate financial reports for the admin.
DeliveryCoordinator	This is a primary human actor who manages the shipment of completed print orders. They coordinate with delivery partners, assign deliveries, update shipment statuses, and ensure customers receive timely tracking information.
PaymentSystem	This is a secondary system actor that integrates with the platform to facilitate online financial transactions. It validates card information, processes payments, and returns confirmation or failure results to the system.
NotificationSystem	This is a secondary system actor responsible for delivering automated messages to customers including order status updates, payment confirmations, and cancellation notices via email or SMS.
SystemTimer	This is a secondary system actor that triggers automated scheduled events such as sending reminders for pending payments or overdue order updates without requiring manual intervention.

ambiguous actor??

B- Task 3.2 — Use Cases Extraction

Lead: Dareen Abualhaj

Contributors: Tala Saabneh (extraction), Razan Froukh (reviewing), Taima Nazal (discussion & editing), Shahd Shwekyeyeh (reviewing)

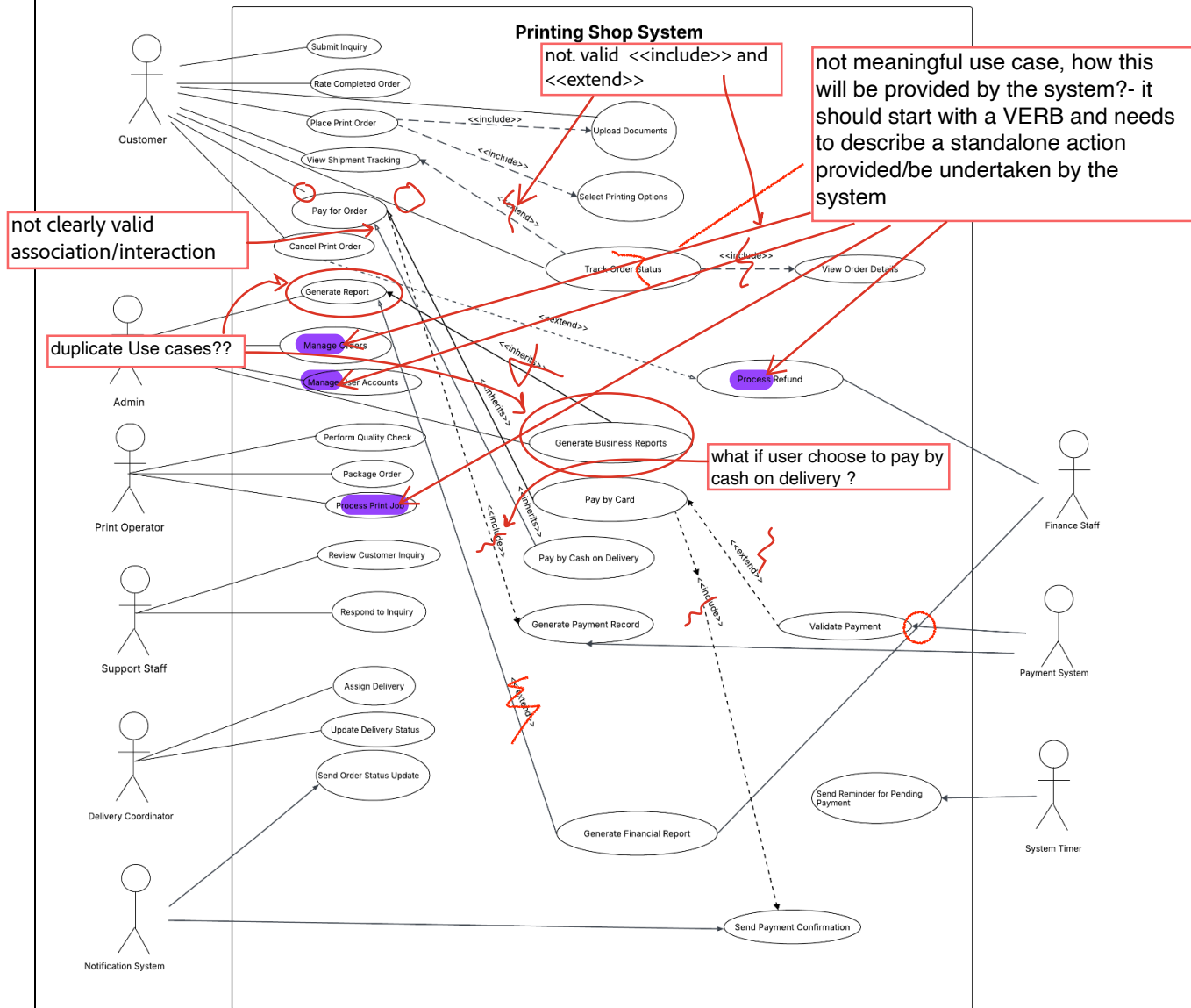


Figure 1 : USE-CASE diagram

The following table identifies all use cases extracted from the system requirements (Phase 2) and business description (Phase 1), organized by actor:

1.1 Customer Use Cases

UC ID	Use Case Title	Actor
UC1	Upload Documents	Customer
UC2	Select Printing Options	Customer
UC3	Place Print Order	Customer
UC4	Pay for Order	Customer
UC4a	Pay by Card	Customer
UC4b	Pay by Cash on Delivery	Customer
UC5	Track Order Status	Customer
UC5a	View Order Details	Customer
UC5b	View Shipment Tracking	Customer
UC6	Cancel Print Order	Customer
UC7	Submit Inquiry	Customer
UC8	Rate Completed Order	Customer

1.2 Admin Use Cases

UC ID	Use Case Title	Actor
UC9	Generate Business Reports	Admin
UC10	Manage Orders	Admin
UC11	Manage User Accounts	Admin

1.3 Print Operator Use Cases

UC ID	Use Case Title	Actor
UC12	Process Print Job	Print Operator
UC13	Perform Quality Check	Print Operator
UC14	Package Order	Print Operator

1.4 Support Staff Use Cases

UC ID	Use Case Title	Actor
UC15	Review Customer Inquiry	Support Staff
UC16	Respond to Inquiry	Support Staff

1.5 Finance Staff Use Cases

UC ID	Use Case Title	Actor
UC17	Process Refund	Finance Staff
UC18	Generate Financial Report	Finance Staff

1.6 Delivery Coordinator Use Cases

UC ID	Use Case Title	Actor
UC19	Assign Delivery	Delivery Coordinator
UC20	Update Delivery Status	Delivery Coordinator

1.7 Secondary Actor Use Cases

UC ID	Use Case Title	Actor
UC4c	Validate Payment	Payment System
UC4d	Generate Payment Record	Payment System
N1	Send Order Status Update	Notification System
N2	Send Payment Confirmation	Notification System
ST1	Send Reminder for Pending Payment	System Timer

1.8 Parent Use Case

UC ID	Use Case Title	Actor
UCR	Generate Report	— (parent)

Task 3.3 — Use Case Specifications

System Name: Printing Shop System

B+

Tala Saabneh – 1221464

Use Case: Place Print Order		Valid use case-Title of use case: Should be a use case as defined in the use case diagram and related your chosen scenario
Use Case Title	Place Print Order	
Purpose	To allow Customers to create and submit print orders by uploading printable documents, selecting printing configurations, choosing payment methods, and confirming the order for processing and fulfilment.	
Description	This use case manages the creation and submission of print orders within the Printing Shop System. The Customer uploads printable documents, selects printing configurations, chooses a payment method, and confirms the order. The system validates uploaded files, calculates printing cost, processes payment transactions, stores order information, and updates order status for fulfilment and tracking purposes.	
Actors	• Primary Actor: • Customer • Secondary Actors: • Payment System • Notification System • Print Operator	valid actors
Data	• Customer account information • Printing preferences • Order details • Order status records • Uploaded document files • Payment information • Delivery information • Transaction logs	
Trigger / Stimulus	The Customer selects the "Place Order" option after uploading documents and selecting printing preferences.	valid trigger
Preconditions	1. The Customer is authenticated and logged into the system. 2. At least one valid printable document has been uploaded successfully. 3. Printing preferences have been selected for all uploaded documents. 4. The system is connected to the order database and payment gateway service. 5. Uploaded files satisfy supported format and size constraints.	Valid pre-conditions, should specify valid state of system on entry Invalid pre-conditions, should specify valid state of system on entry
Workflow / Sequence of Events (Normal Flow)		
Normal Flow	1. The Customer accesses the print ordering section. 2. The system displays the document upload interface and supported file formats. 3. The Customer uploads one or more printable document files. 4. The system validates uploaded files and confirms successful upload. 5. The system displays available printing options for each uploaded document. 6. The Customer selects printing preferences, including paper size, print color, number of copies, and printing quality. 7. The system validates the selected printing preferences and calculates the total cost.	GOOD-Reasonable-Needs to be a sequence of FLOW? Add more specific details of the exact follow to match with the business scenario, Add alternative and error follows, using conditional operators. Steps Should be numbered

	8. The system displays the order summary, including uploaded documents, selected preferences, quantities, and total estimated cost. 9. The Customer reviews the order summary and selects a payment method: Online Card Payment or Cash on Delivery. 10. If Online Card Payment is selected, the system sends a payment request to the Payment System, and the Customer enters payment details. 11. The Payment System validates and authorizes the transaction, and the system receives the authorization result. 12. The system generates a unique order identifier and stores the order and document information in the database. 13. The system updates the order status to "Paid" for successful online payment or "Pending Payment" for Cash on Delivery. 14. The system records transaction details and order timestamps. 15. The Notification System sends an order confirmation notification to the Customer, and the system notifies the Print Operator about the newly submitted print order. 16. The system displays a successful order confirmation message.
Alternative Flows	
A1 – Cash on Delivery Payment	1. The Customer selects "Cash on Delivery" as the payment method. 2. The system skips online payment authorization processing. 3. The system records the order with status "Pending Payment." 4. The order proceeds to fulfilment and delivery processing. Add appropriate conditional operator (IF) in the flow
A2 – Recommended Printing Preferences	1. After document upload, the system analyzes document properties. 2. The system recommends predefined printing settings suitable for the uploaded document. 3. The Customer accepts the recommended printing preferences. 4. The workflow resumes from Step 7 of the Main Flow. Add appropriate conditional operator (IF) in the flow
A3 – Customer Modifies Printing Preferences	1. The Customer reviews the order summary before confirmation. 2. The Customer modifies one or more printing preferences. 3. The system recalculates the printing cost. 4. The updated order summary is displayed. 5. The workflow resumes from Step 10 of the Main Flow. Add appropriate conditional operator (IF) in the flow
Error Flows	
E1 – Unsupported File Format	1. The Customer uploads a file with an unsupported format. 2. The system rejects the uploaded file. 3. The system displays: "Unsupported file format. Please upload PDF, DOCX, or JPG files only." 4. The workflow is suspended until a supported file is uploaded.
E2 – Online Payment Failure	1. The Customer selects Online Card Payment. 2. The Payment System fails to authorize the transaction. 3. The system displays a payment failure notification. 4. The system temporarily preserves the order information. 5. The Customer may retry payment or select another payment method. 6. The failed transaction attempt is recorded in the transaction log.
E3 – Database Failure During	1. The Customer confirms the print order. 2. The system fails to store order information due to database timeout or connection failure. 3. The system cancels the order transaction. Not Valid error- system error

Order Processing	- The system displays: "System Error: Unable to process your order at this time." - No order record is committed to the database. - The system records the failure event in the system log.
E4 – Missing Printing Preferences Not Valid error- system error	- The Customer uploads documents without selecting complete printing preferences. - The system detects incomplete printing configuration. - The system displays a validation message requesting completion of all required printing preferences. The order process is suspended until all required options are selected.
Postconditions	
Successful Completion	- The print order is successfully stored in the system database. - A unique order identifier is assigned to the order. - Uploaded documents are associated with the generated order identifier. - The order status is updated appropriately: "Paid" or "Pending Payment". - The order is added to the Printing Staff processing queue. - Payment transaction details are securely recorded. - Order tracking becomes available through the Customer account. - Order creation and payment events are logged for auditing purposes.
Unsuccessful Completion	- No incomplete order information is committed to the database. - Invalid upload and payment transactions are securely rejected. - Database consistency is maintained. - Failed transactions and processing attempts are recorded in the system log.
Business Rules / Constraints	
Business Rules	- Only authenticated Customers are allowed to place print orders. - Uploaded documents must satisfy supported file format and size constraints. - Successful online payment authorization is required before order status is marked as "Paid." - Each print order must have a unique order identifier. - All order status changes must be recorded with timestamps for tracking and auditing purposes. - The system shall maintain order consistency during simultaneous order submissions.
Comments	- All payment communication with the Payment System must be secure and encrypted. - The system shall continuously validate uploaded documents and printing configurations during order creation. - The system should automatically notify Customers whenever order status changes occur. - All order processing activities must be logged for operational monitoring and auditing purposes. - The system shall ensure atomic transaction processing to prevent partial order creation during failures.

C+

Dareen Abualhaj - 1220686

System Name	Printing Shop System	Valid use case-Title of use case: Should be a use case as defined in the use case diagram and related your chosen scenario
Use Case Title	Upload Documents	
Description	A Customer uploads one or more documents to the system in preparation for placing a print order. The system validates each uploaded file for format and size, associates it with the customer's current order session, and confirms successful submission. The customer must upload at least one valid document before proceeding to select printing options.	
Actors	Primary: Customer	valid actors
Data	Document file(s) (PDF, DOCX, JPEG, PNG), file size, file format, customer session ID, file submission time	
Stimulus / Trigger	The Customer selects the upload option and submits one or more files to the system.	POOR valid trigger
Pre-conditions	1. The Customer has an active account and is currently logged into the system. 2. The Customer has at least one document available on the system to upload.	Valid pre-conditions, should specify valid state of system on entry
Workflow OR Sequence/Flow of Events	Normal Flow: 1. The Customer navigates to the document upload section. 2. The Customer clicks 'Browse' and selects a file from their device. 3. The system checks the file format against the supported types (PDF, DOCX, JPEG, PNG). 4. The system checks that the file size does not exceed the maximum allowed limit. 5. The system accepts the file and associates it with the Customer's current order session. 6. The system displays the uploaded file name, format, and size as a confirmation to the Customer. 7. The Customer clicks 'Confirm Upload'. 8. The system stores the document temporarily and links it to the order session ID. 9. The Customer proceeds to select printing options . Alternative Flow A1 – Upload via Drag and Drop: At step 2, instead of clicking 'Browse', the Customer drags and drops one or more files directly into the upload area. The system receives the file(s) and continues from step 3 of the normal flow.	

POOR VALID- Needs to be a sequence of FLOW? Add more specific details of the exact follow to match with the business scenario, Add alternative and error follows, using conditional operators. Steps Should be numbered

POOR VALID. Add appropriate conditional operator (IF) in the flow

	Alternative Flow A2 – Upload from Cloud Storage Add appropriate conditional operator (IF) in the flow At step 2, the Customer selects the 'Upload from Cloud' option to a supported cloud service (e.g., Google Drive). The Customer selects a file from their cloud storage. The system downloads and receives the file, then continues from step 3 of the normal flow. Error Flow E1 – Unsupported File Format: At step 3, the system detects that the uploaded file format is not supported. The system rejects the file and displays: 'Unsupported file format. Please upload a file in PDF, DOCX, JPEG, or PNG format.' The Customer is prompted to select a valid file. No file is stored. Error Flow E2 – File Size Exceeded: At step 4, the system detects that the uploaded file exceeds the maximum allowed size. The system rejects the file and displays: 'File size exceeded. Please upload a file within the allowed size limit.' The Customer is prompted to upload a smaller or compressed version. No file is stored. Error Flow E3 – Empty Submission Attempt: If the Customer attempts to proceed to printing options without uploading any document, the system detects that no file is associated with the session. The system displays: 'At least one document must be uploaded before proceeding.' The Customer is redirected back to the upload section. Not Valid error- system error Error Flow E4 – Upload Failure Due to System Error: At step 8, the system fails to store the uploaded file due to a server or database error. The system displays: 'Upload failed. Please try again.' No file is saved and the session remains active. The Customer can retry. Not Valid error- system error
Post-conditions / Response INVALID post-conditions, should specify valid state of system on exit/end of use case	1. All validated documents are temporarily stored in the system and linked to the Customer's active order session ID. VALID post-conditions, should specify valid state of system on exit/end of use case 2. The Customer's session state is updated once a document has been successfully uploaded. 3. Any rejected files are not stored in the system. 4. The system is ready to present printing options for the uploaded document(s).
Comments	1. Supported formats are: PDF, DOCX, JPEG, PNG. 2. Maximum file size per document is defined by the system administrator. 3. Uploaded files are temporarily stored until the order is confirmed or the session expires.

C-

Razan Froukh -1222192

System Name	Printing Shop System	Ambiguous/Valid use case-Title of use case: Should be a use case as defined in the use case diagram and related your chosen scenario
Use Case Title	Track Order Status	
Purpose	To allow Customers to monitor and track the progress of their print orders and shipment updates in real time.	
Description	This use case allows the Customer to monitor the progress and current status of a submitted print order. The system provides real-time updates regarding order processing, printing, packaging, shipment, and delivery stages. Customers can also view shipment tracking information once the order is dispatched.	
Actors	Primary Actor: Customer Secondary Actors: Notification System, Delivery Coordinator	invalid actors
Data	• Order ID, Customer account information, Order status records, Shipment tracking information, Delivery updates, Notification records	
Stimulus / Trigger	The Customer selects the "Track Order Status" option from their account dashboard.	valid trigger
Pre-conditions	1. The Customer is authenticated and logged into the system. 2. The Customer has previously placed at least one print order 3. The requested order exists in the system database. 4. The order has not been permanently deleted or archived.	Valid pre-conditions, should specify valid state of system on entry
Workflow (Normal)	1. The Customer accesses the "My Orders" section. 2. The system retrieves all orders associated with the Customer account. 3. The Customer selects a specific order to track. 4. The system retrieves the latest order status information from the database. 5. The system displays order details including order number, uploaded documents, selected printing options, payment status, and estimated delivery date. 6. The system displays the current processing stage such as "Pending," "Printing," "Quality Check," "Packaged," "Shipped," or "Delivered." 7. If the order has been shipped, the system displays shipment tracking information and courier details. 8. The Customer reviews the displayed order progress information. 9. The system records the access event for monitoring and auditing purposes.	
Alternative Flow A1 – Shipment Tracking Available	1. The Delivery Coordinator updates shipment information. 2. The system receives shipment tracking details.	Add appropriate conditional operator (IF) in the flow

Invalid Alternative Flow

	3. The system displays the live shipment tracking link and estimated arrival time. 4. The workflow resumes from Step 8 of the Main Flow.
Alternative Flow A2 – Order Delivered Successfully Invalid Alternative Flow	1. The system detects that the order status is “Delivered.” 2. The system displays delivery confirmation details and delivery timestamp. 3. The Customer <u>may proceed</u> to rate the completed order.
Error Flow E1 – Invalid Order Selection 	1. The <u>Customer selects an invalid or unavailable</u> conflict with pre conditions 2. The system fails to retrieve order details. 3. The <u>system displays: “Selected order could not be found.”</u> 4. The Customer is redirected back to the orders list. Not Valid error- system error
Error Flow E2 – Database Connection Failure Not Valid error- system error	1. The system attempts to retrieve order status information. 2. The database connection fails or times out. 3. The system displays: “Unable to retrieve order status at this time. Please try again later.” 4. The failure event is recorded in the system log.
Error Flow E3 – Shipment Information Unavailable	1. The order has been shipped but shipment tracking data is temporarily unavailable. 2. The system displays: “Shipment tracking information is currently unavailable.” poor valid error 3. The Customer <u>may retry</u> later.
Post-conditions / Response (Successful Completion)	1. The Customer successfully views the latest order status. 2. Shipment tracking information is displayed when available. 3. <u>Order access activity is recorded in the system logs.</u> 4. <u>The Customer remains informed about order progress and delivery updates.</u>
Post-conditions / Response (Unsuccessful Completion) 	1. No changes are made to order information. 2. System errors and failed retrieval attempts are recorded in sy 3. Database consistency remains maintained. INVALID post-conditions, should specify valid state of system on exit/end of use case
Business Rules / Constraints	1. Only authenticated Customers can access their own orders. 2. Customers cannot access order information belonging to other users. 3. Order status updates must be synchronized with printing and delivery operations. 4. Shipment tracking information must only appear after shipment assignment.

	5. All order status changes must be timestamped for auditing purposes.
Comments	1. The system should automatically refresh order progress updates periodically. 2. Customers should receive notifications whenever order status changes occur. 3. Shipment tracking integration should support external courier services. 4. All tracking activities should be logged for security and monitoring purposes.

C+

Shahd Shwekeyeh – 1222105

Use Case : Cancel Print Order	
System Name	Printing Shop System
Use Case Title	Cancel Print Order
Purpose	To allow authenticated Customers to request the cancellation of a placed print order, provided that the order has not yet entered the processing stage. The system verifies the order status, updates it accordingly, and initiates a refund if the order was paid online.
Description	This use case handles the cancellation of a print order by the Customer. The Customer selects a pending order and submits a cancellation request. The system verifies that the order is still in a cancellable state (i.e., status is Pending or Pending Payment). Upon successful verification, the system updates the order status to Cancelled and sends a confirmation notification. If the order was paid online, the system automatically initiates a refund through the Finance Staff workflow.
Actors	Primary Actor: Customer Secondary Actors: Notification System ,Finance Staff
Data	• Customer account information , Order ID and order details ,Order status, Payment method and transaction record , Cancellation timestamp , Refund record (if applicable)
Stimulus / Trigger	The Customer selects the "Cancel Order" option from their order details page.

Pre-conditions	
Pre-conditions	1. The Customer is authenticated and logged into the system 2. The selected order exists in the system and belongs to the Customer. 3. The order status is "Pending" or "Pending Payment" (i.e., has not entered processing). 4. The system is connected to the order database.
Workflow / Sequence of Events	
Normal Flow	1. The Customer navigates to the "My Orders" section. 2. The system retrieves and displays all orders associated with the Customer account. 3. The Customer selects a specific order to view its details. 4. The system displays the order details including status, documents, and payment information. 5. The Customer clicks the "Cancel Order" button. 6. The system checks that the order status is "Pending" or "Pending Payment". 7. The system displays a cancellation confirmation prompt: "Are you sure you want to cancel this order?" 8. The Customer confirms the cancellation request. 9. The system updates the order status to "Cancelled" and records the cancellation timestamp. 10. The Notification System sends a cancellation confirmation message to the Customer. 11. If the order was paid online, the system flags the order for refund processing by Finance Staff. 12. The system displays a cancellation success message to the Customer.
Alternative Flows	
A1 – Customer Cancels Before Payment	1. The Customer selects an order with status "Pending Payment" (no online payment made). 2. The system processes the cancellation without initiating a refund. 3. The order status is updated to "Cancelled". 4. The workflow resumes from Step 10 of the Normal Flow.

Valid pre-conditions, should specify valid state of system on entry

Invalid pre-conditions, should specify valid state of system on entry

REASONABLE- Needs to be a sequence of FLOW? Add more specific details of the exact follow to match with the business scenario, Add alternative and error follows, using conditional operators. Steps Should be numbered

Add appropriate conditional operator (IF) in the flow

A2 – Customer Decides Not to Cancel

Invalid Alternative Flow

1. At Step 8, the Customer selects "No" on the confirmation prompt.
2. The system cancels the cancellation request.
3. The order status remains unchanged.
4. The Customer is redirected to the order details page.

Error Flows

E1 – Order Already in Processing

Invalid Error Flow.
Conflict with pre conditions

1. At Step 6, the system detects that the order status is "Processing", "Printed", "Shipped", or "Delivered".
2. The system rejects the cancellation request.
3. The system displays: "This order cannot be cancelled as it has already entered the processing stage."
4. The Customer is redirected to the order details page.

E2 – Database Failure During Cancellation

Not Valid error- system error

1. At Step 9, the system fails to update the order status due to a database error.
2. The system rolls back any partial changes.
3. The system displays: "Unable to process your cancellation at this time. Please try again later."
4. The failure event is recorded in the system log.

E3 – Session Timeout During Cancellation

Not Valid error- system error

1. The Customer session expires before confirming the cancellation request.
2. The system discards the cancellation request.
3. The Customer is redirected to the login page.
4. No changes are made to the order.

Post-conditions / Response

Successful Completion

1. The order status is updated to "Cancelled" in the system database.
2. The cancellation timestamp is recorded with the order.
3. The Customer receives a cancellation confirmation notification.
4. If applicable, a refund request is initiated and assigned to Finance Staff.
5. The cancellation event is logged for auditing purposes.

Unsuccessful Completion

INVALID post-conditions,
should specify valid state
of system on exit/end of
use case

1. No changes are made to the order status.
2. Failed cancellation attempts are recorded in the system logs.
3. Database consistency is maintained.

Business Rules / Constraints**Business Rules**

1. Only authenticated Customers can request order cancellation.
2. Cancellation is only permitted when the order status is "Pending" or "Pending Payment".
3. Orders that have entered processing, printing, shipping, or delivery cannot be cancelled.
4. A refund must be initiated for any order that was paid online and subsequently cancelled.
5. All cancellation events must be timestamped and logged for auditing purposes.

Comments**Comments**

- The system should send cancellation notifications via email or SMS through the Notification System.
- Refund processing timelines are subject to the payment provider's policies and are handled by Finance Staff.
- The system should prevent double-cancellation requests for an already-cancelled order.
- All cancellation and refund activities must be logged for operational monitoring and compliance.

System Name : Online Printing System	
Use Case Title	SELECT PRINTING OPTIONS
Purpose	To allow the Customer to configure printing preferences for each uploaded document before order confirmation, ensuring accurate cost calculation and correct print job execution.
Description	This use case manages the selection and validation of printing preferences within the Printing Shop System. After uploading documents, the Customer selects printing options including paper size, color mode, page orientation, number of copies, and print quality for each document. The system validates the selected options, calculates the estimated printing cost, and confirms the configuration before allowing the Customer to proceed to order placement.
Actors	Primary Actor: Customer Secondary Actors: Notification System
Data	Uploaded document files, Selected printing preferences (paper size, color mode, orientation, copies, quality), Estimated printing cost, Customer session ID, Order configuration records
Stimulus / Trigger	The Customer selects the "Configure Printing Options" step after successfully uploading one or more documents.
Pre-conditions	1. The Customer is authenticated and logged into the system. 2. At least one valid document has been successfully uploaded to the current order session. 3. The system printing options interface is accessible and operational. 4. The system is connected to the order configuration database.
Workflow / Sequence of Events	
Notes	1. The Customer navigates to the printing options configuration section. 2. The system retrieves the list of uploaded documents associated with the current order session. The system displays the printing options interface for the first uploaded document. The Customer selects paper size from the available options (e.g., A4, A3, Letter). The Customer selects color mode: Black & White or Full Color.

Valid use case-Title of use case: Should be a use case as defined in the use case diagram and related your chosen scenario

a sub-use case?
Choose a main use case?

invalid actors

valid trigger

Valid pre-conditions, should specify valid state of system on entry

invalid pre-conditions, should specify valid state of system on entry

REASONABLE- Needs to be a sequence of FLOW? Add more specific details of the exact follow to match with the business scenario, Add alternative and error follows, using conditional operators. Steps Should be numbered

	- The Customer selects page orientation: Portrait or Landscape. - The Customer specifies the number of copies (minimum: 1). - The Customer selects print quality: Standard, High, or Draft. - The system validates the selected printing options for the current document. - The system calculates and displays the estimated cost for the current document based on the selected options. - If multiple documents are uploaded, the system prompts the Customer to configure options for the next document; steps 4–10 are repeated for each document. - The system displays a summary of printing options and estimated costs for all uploaded documents. - The Customer reviews and confirms the selected printing options. - The system saves the confirmed printing preferences and associates them with the current order session. - The Customer proceeds to order placement.
Alternative Flows	
Add appropriate conditional operator (IF) in the flow A1 – Apply Same Options to All Documents	At Step 3, the Customer selects the option "Apply same printing options to all documents". - The system displays the printing options interface once, applying selections globally to all uploaded documents. - The Customer configures the printing preferences. - The system validates and applies the selected options to all uploaded documents. - The workflow resumes from Step 12 of the Normal Flow.
Add appropriate conditional operator (IF) in the flow A2 – System Recommends Printing Options	After Step 3, the system analyzes the uploaded document properties. - The system recommends predefined printing settings suitable for the document (e.g., A4, Black & White, Portrait, Standard Quality). - The Customer accepts the recommended printing options. - The system applies the recommended settings and calculates the estimated cost. - The workflow resumes from Step 12 of the Normal Flow.
Add appropriate conditional operator (IF) in the flow A3 – Customer Modifies Previously Saved Options	At Step 12, the Customer selects a document to modify its printing preferences. - The system displays the current saved options for the selected document. - The Customer updates one or more printing preferences.

	- The system recalculates the estimated cost for the modified document. - The updated summary is displayed. - The workflow resumes from Step 13 of the Normal Flow.
Error Flows	
E1 – Invalid Number of Copies Not Valid error- system error	- At Step 7, the Customer enters an invalid value for the number of copies (e.g., 0, negative number, or non-integer). - The system rejects the input and displays: "Invalid number of copies. Please enter a positive whole number." - The Customer is prompted to re-enter a valid number of copies. - No options are saved until a valid value is provided.
Not Valid error- system error E2 – Incomplete Options Selection	- At Step 9, the system detects that one or more required printing options have not been selected. - The system highlights the missing fields and displays: "Please complete all required printing options before proceeding." - The Customer is prompted to fill in the missing selections. - The order process is suspended until all required options are configured.
E3 – Cost Calculation Failure	- At Step 10, the system fails to calculate the estimated cost due to a server or configuration error. - The system displays: "Unable to calculate printing cost at this time. Please try again." - The selected options remain temporarily stored in the session. - The Customer may retry or contact customer support. - The failure event is recorded in the system log. Not Valid error- system error
E4 – Session Timeout During Configuration	- The Customer's session expires while configuring printing options. - The system discards all unsaved printing option selections. - The system displays: "Your session has expired. Please log in again to continue." - The Customer is redirected to the login page. - No printing preferences are saved.
Post-conditions / Response	

Successful Completion	1. Printing preferences are saved and associated with each uploaded document in the order session. 2. Estimated printing cost is calculated and displayed to the Customer. 3. The order session is updated to reflect that printing options have been fully configured. 4. The system is ready to proceed to the order placement stage. 5. Printing configuration activity is recorded for auditing purposes.
Unsuccessful Completion	1. No incomplete or invalid printing configuration is saved to the system. 2. Failed configuration attempts are recorded in the system logs. INVALID post-conditions, should specify valid state of system on exit/end of use case Customer's session remains active to allow retry. base consistency is maintained.
Business Rules / Constraints	
Business Rules	1. Only authenticated Customers may configure printing options. 2. Printing options must be fully configured for every uploaded document before order confirmation. 3. The number of copies shall be a positive integer of at least 1. 4. Paper size selection shall be limited to the sizes supported by the printing shop. 5. The estimated cost shall be automatically recalculated whenever printing options are modified. 6. Printing preferences shall be associated with the corresponding document identifier and order session.
Comments	1. The system should support at minimum the following paper sizes: A4, A3, and Letter. 2. The system should clearly display the cost impact of each printing option change to assist Customers in decision-making. 3. Recommended printing options (SR2.6) should be based on predefined document rules configured by the Admin. 4. All printing option configuration activities should be logged for operational monitoring. 5. The system should allow the Customer to preview estimated total cost across all uploaded documents before proceeding.

B- Task 3.5- Activity Modelling

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GOOD valid use
of swim-lane - activity flow
must show interactions
between different actors

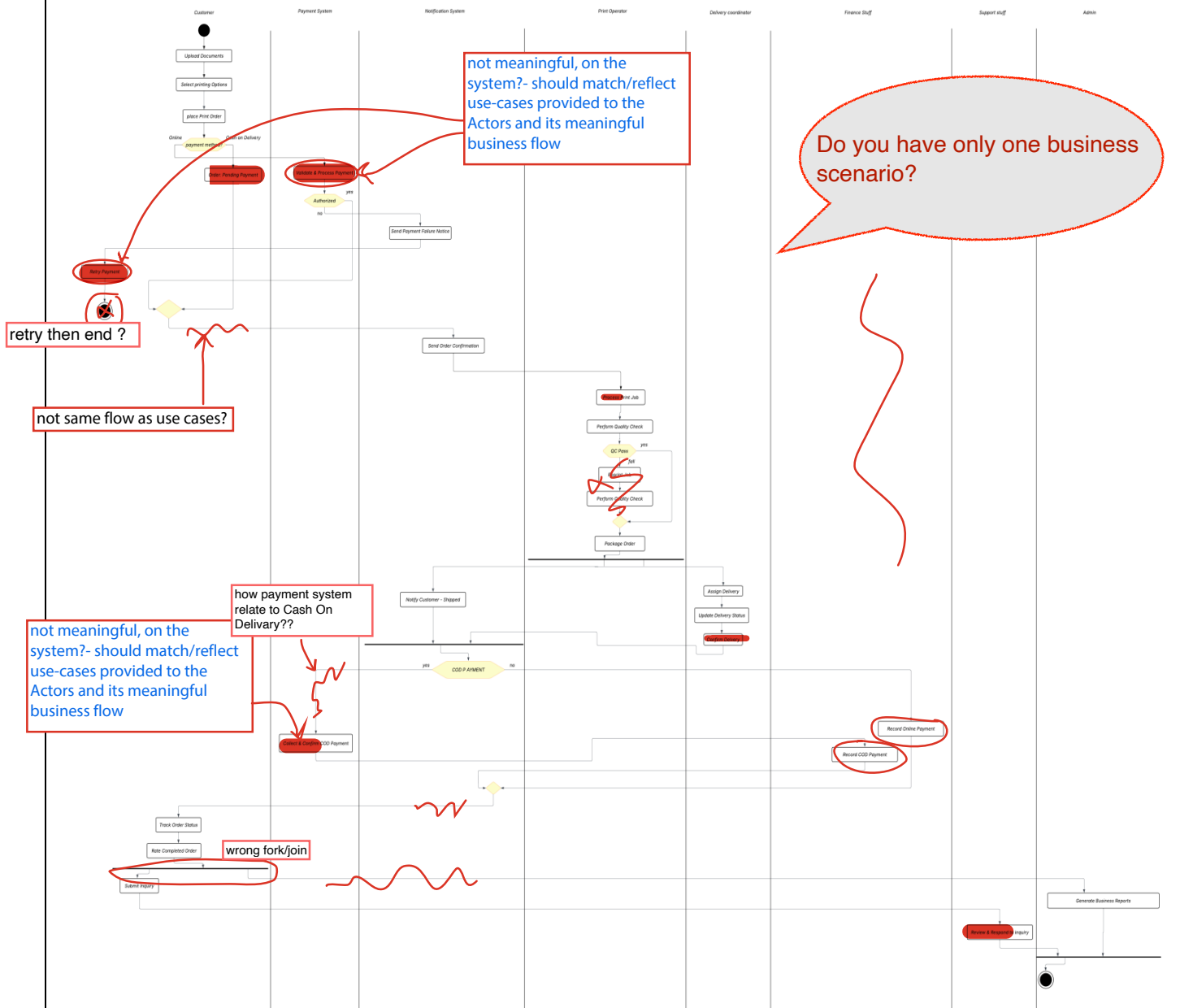


Figure2 :Main ACTIVITY diagram

Designs links:

➔ **USE-CASE diagram**

➔ **ACTIVITY diagram**